

COS30082

Applied Machine Learning



Lecture 2

Overview of Machine Learning Algorithms

2.0 Algorithm Overview

Types of Machine Learning

House Age	Bedrooms	Size (sqm)	Price (\$1000)
5	3	120	420
10	4	150	500
2	2	90	350

■ Supervised Learning

- Learn from data with known correct answers.
- Example: Learn to predict house prices from past house data.

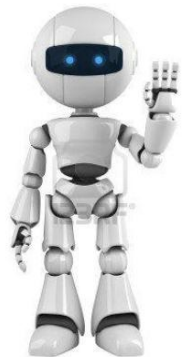
■ Unsupervised Learning

- Learn from data without known answers to find patterns or groups in the data.
- Example: Learn to group customers with similar behavior.



■ Reinforcement Learning

- Learn by trying actions, receiving rewards or penalties, and improving decisions over time.
- Example: A robot learns to navigate an environment by trying actions, receiving rewards or penalties, and improving over time.



Types of Machine Learning

Some variations:

- **Semi-Supervised Learning**

- Use a small amount of data with correct labels and a large amount of data without labels to train the model.
- Steps: Train on labeled data → predict unlabeled data → select high-confidence predictions → add to training data → retrain and repeat.

- **Self-Supervised Learning**

- The model creates its own labels from the data.
- Example: Mask (hide) words in a sentence and train the model to predict the missing words. The correct answers are taken from the data itself.

Original sentence:

The quick brown fox jumps over the lazy dog.

Masked sentences:

The quick [MASK] fox jumps over the lazy dog.

The quick brown fox [MASK] over the [MASK] dog.

Machine Learning Tasks

Supervised Learning Tasks

- Regression: Predict a number (e.g., house price).
- Classification: Predict a category (e.g., spam or not, which animal).
- Ranking: Predict order (e.g., ranking search results).
- Sequence Prediction: Predict a sequence (e.g., language translation).

Unsupervised Learning Tasks

- Clustering: Group similar data points without known labels (e.g., group customers based on purchasing behavior).
- Association Rule Mining: Discover relationships between items in large datasets (e.g., customers who buy bread often also buy milk).
- Dimensionality Reduction: Reduce the number of features while keeping important information (e.g., reduce 100 features to 2 features for visualization).

Reinforcement Learning Tasks

- Decision Making – choose the best action (e.g., game playing)
- Control – control a system (e.g., robot navigation)

Machine Learning Algorithms

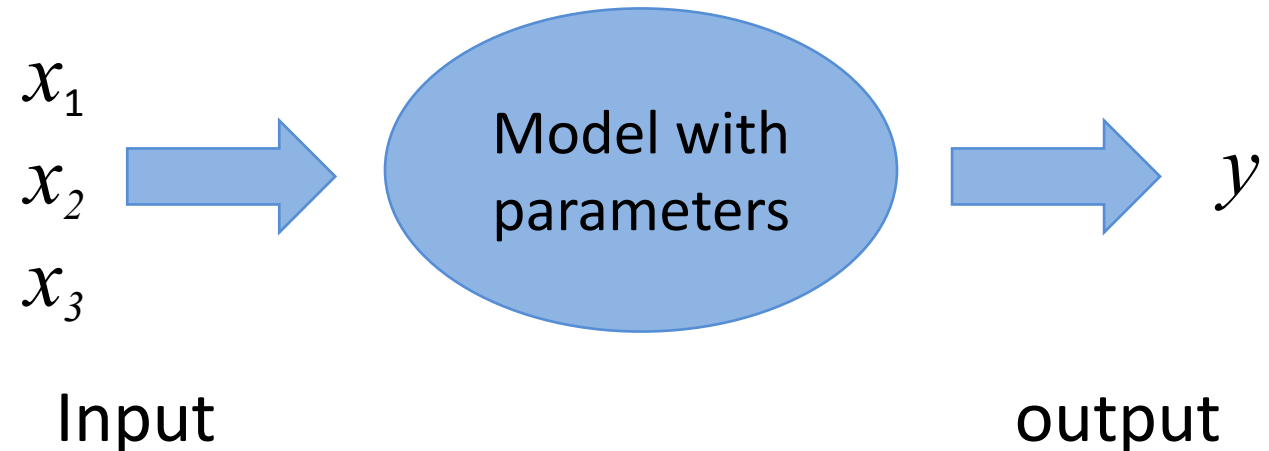
- Linear Regression — Predict a continuous value using a linear relationship between variables.
- Logistic Regression — Predict the probability of a class for classification tasks.
- Decision Tree — Make predictions using a tree of decision rules.
- Support Vector Machine (SVM) — Find the optimal boundary that separates classes.
- K-Nearest Neighbors (KNN) — Predict based on the most similar nearby data points.
- K-Means Clustering — Group similar data points into clusters.
- Neural Networks — Learn complex patterns through layers of interconnected nodes.

Linear Regression

In supervised learning, many algorithms follow a similar training idea:

- They assume a model (a function) with parameters.
- The model is trained using labeled data. The parameters are adjusted to minimize prediction error.
- Once the parameters are learned, the model can be used to make predictions.

x_1	x_2	x_3	y
2	5	1	18
3	6	2	23
4	7	1	24
5	8	3	31



Linear Regression

- **Model Assumption:** In linear regression, we assume the model is a linear equation.

$$y = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

- **Prediction:** For each training data point $(x_1, x_2, \dots, x_n)$, the model predicts:

$$\hat{y} = w_1x_1 + w_2x_2 + \dots + w_nx_n + b$$

Here $\hat{y}$ is the predicted value.

- **Loss Function:** Linear regression learns the parameters by minimizing the total squared error across all training data.

This is called the least squares method.

- **Training goal:** The goal of training is to find parameters $w_1, w_2, \dots, w_n, b$ that minimize the loss function.

$$L = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Linear Regression

How to minimize the loss?

- Normal Equation

This method computes the exact optimal parameters directly using a mathematical formula. However, it becomes slow when the number of features is very large.

- Gradient Descent

This is an iterative method, and it produces approximate values of the parameters.

- Start with random parameters
- Compute the loss
- Adjust the parameters slightly to reduce the loss
- Repeat many times until the loss becomes small. Update rule:

$$w = w - \alpha \frac{\partial L}{\partial w}$$

where α is the learning rate, and L is the loss function.

Logistic Regression

■ Model Assumption

In logistic regression, we assume the model is a linear equation followed by a sigmoid function:

$$z = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

$$\hat{y} = \sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid function converts the result into a probability between 0 and 1, so we can use logistic regression for binary classification.

■ Prediction

For each training data point $(x_1, x_2, \dots, x_n)$, the model predicts:

$$\hat{y} = P(y = 1 | x)$$

This represents the probability that the input belongs to class 1.

Logistic Regression

- Loss function (Cross-Entropy Loss)

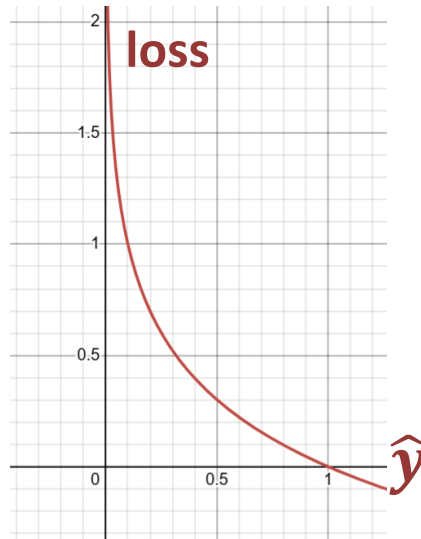
The prediction is compared with the true label y (usually 0 or 1).

$$L = - \sum_{i=1}^n [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)]$$

For each data point,

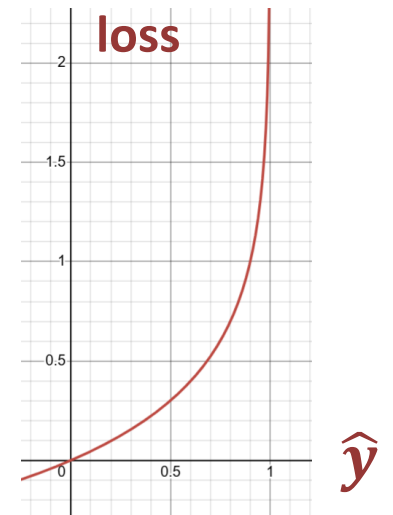
When the true label $y = 1$: loss = $-\log(\hat{y})$

If $\hat{y}$ close to 1, the loss is small
If $\hat{y}$ close to 0, the loss increases rapidly, strongly penalising the model.



When $y = 0$: loss = $-\log(1 - \hat{y})$

If $\hat{y}$ close to 0, the loss is small
If $\hat{y}$ close to 1, the loss increases rapidly.

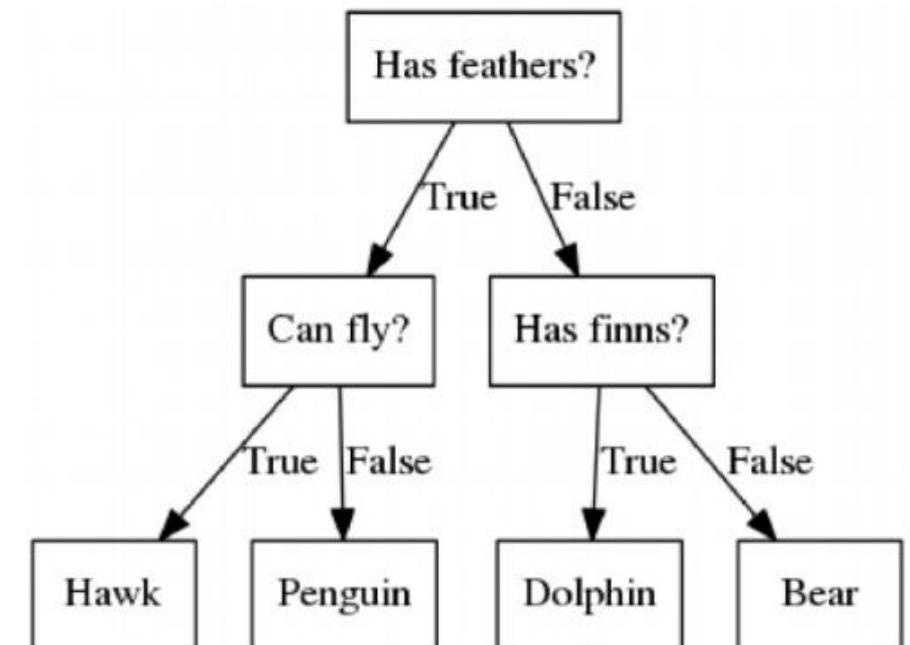
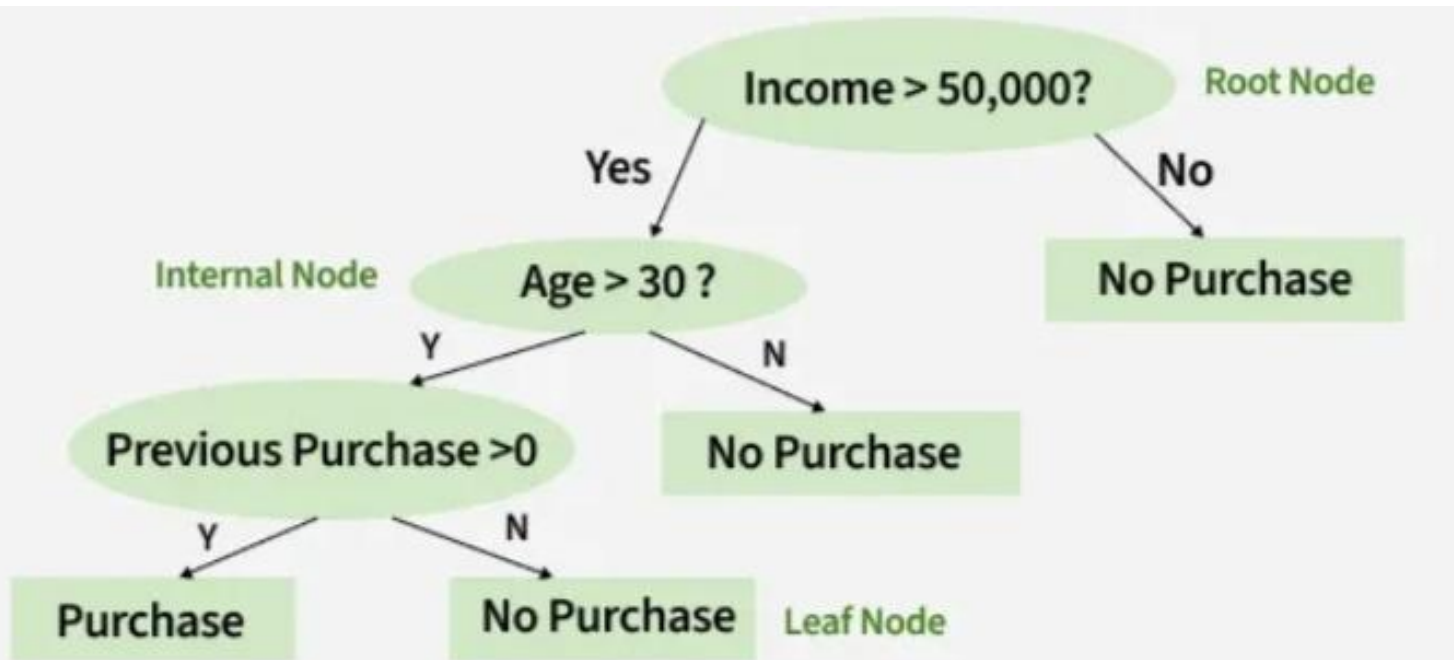


- Optimisation

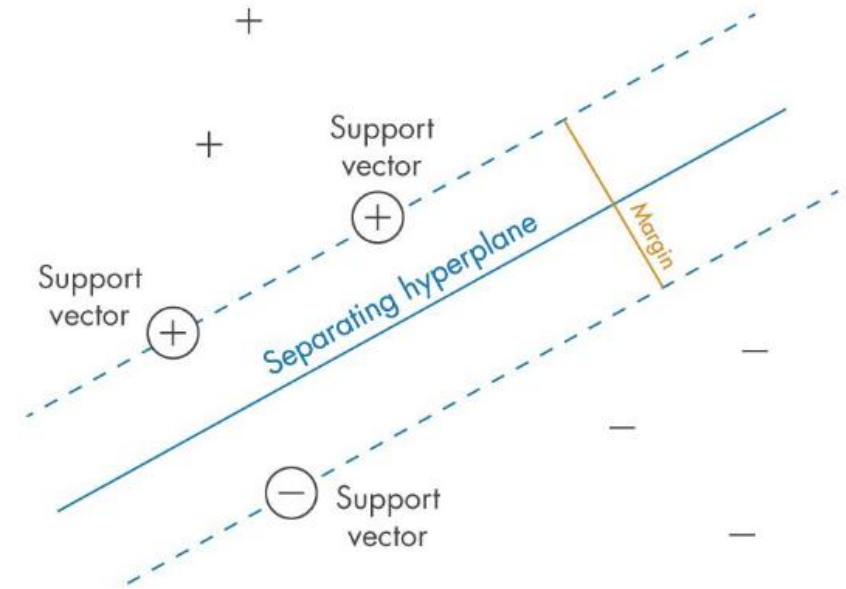
The parameters are usually learned using gradient descent.

Decision Tree

1. Start with all training data.
2. Find the best feature and rule to split the data into smaller groups.
3. Repeat the process for each group.
4. Stop when the groups are pure or small enough.
5. Assign a prediction to each final group (leaf node).



- **Key idea:** SVM tries to find a decision boundary that separates the classes with the largest margin. The margin is the distance between the boundary and the closest data points. These closest points are called support vectors.



- **Model assumption:** SVM assumes the model is a linear decision boundary:

$$w_1x_1 + w_2x_2 + \cdots + w_nx_n + b = 0$$

This equation defines a hyperplane that separates the classes.

- **Prediction:** For an input x , the model computes:

$$z = w_1x_1 + w_2x_2 + \cdots + w_nx_n + b$$

If $z > 0$ predict class 1, otherwise predict class -1.

SVM

- **Hinge Loss:** SVM uses the hinge loss:

$$L = \max(0, 1 - y(w_1x_1 + w_2x_2 + \dots + w_nx_n + b))$$

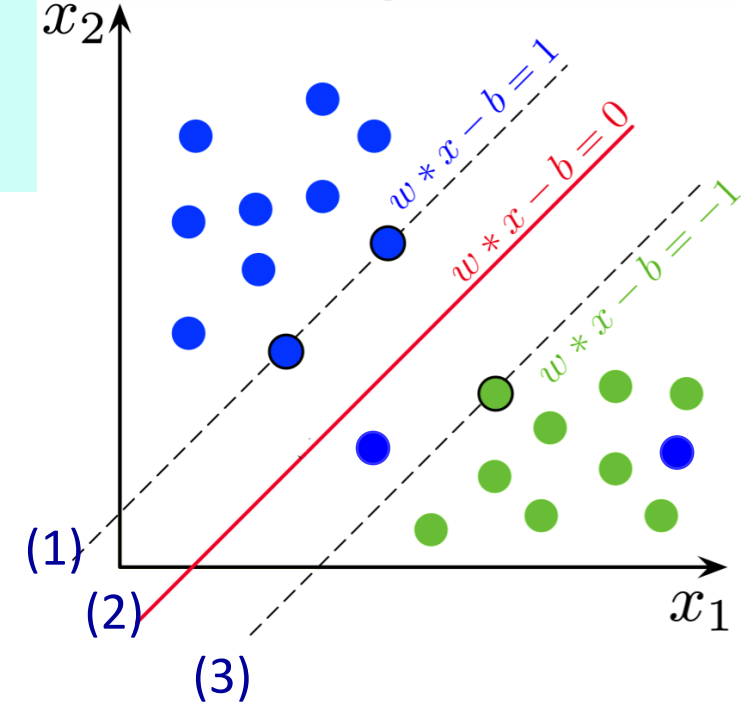
If a point is correctly classified with enough margin,
the loss is 0

If it is misclassified or too close to the boundary, loss > 0

- **SVM Loss Function** $L = \frac{1}{2}(w_1^2 + w_2^2 + \dots + w_n^2) + C \sum_{i=1}^m L_i$

Where:

- First term: for maximizing the margin
(the first term is $\frac{1}{2} \|w\|^2$, the margin is $\frac{1}{\|w\|}$)
- Second term: hinge loss, for reducing classification errors.
- C controls the trade-off between margin and classification error

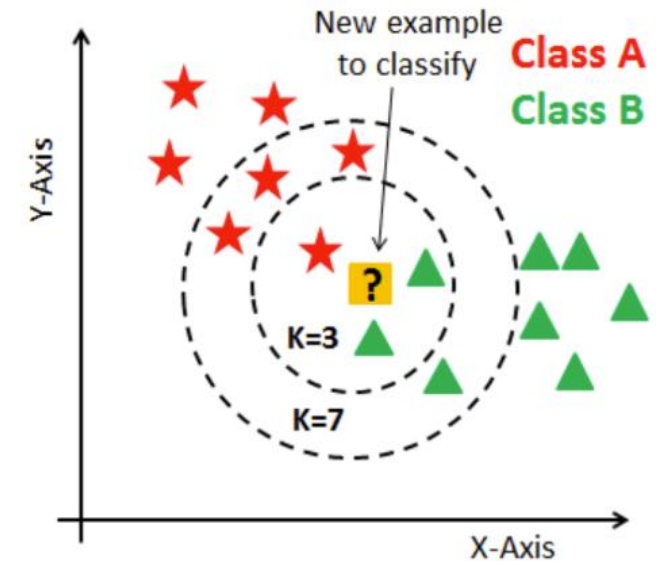


For example, blue dots on or above line (1) have zero hinge loss.

If they fall below line (1), the loss becomes positive. The farther below the line, the larger the loss.

K Nearest Neighbours (KNN)

- **Idea:** Similar inputs tend to have similar outputs. KNN predicts the output based on the most similar training examples.
- **Steps:**
 - Choose a value for K (number of neighbors).
 - Compute the distance between the new data point and all training data
 - Find the K nearest neighbors.
 - Make prediction based on these neighbors.
- **Prediction**
 - Classification: choose the most common class among the neighbors.
 - Regression: take the average value of the neighbors.



If K=3, the new example is classified as Class B.
If K=7, the new example is classified as Class A.

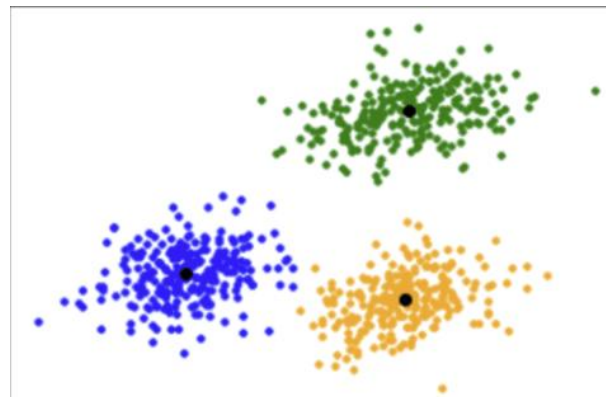
K Means Clustering

- **Idea**

K-means groups data into K clusters based on similarity.

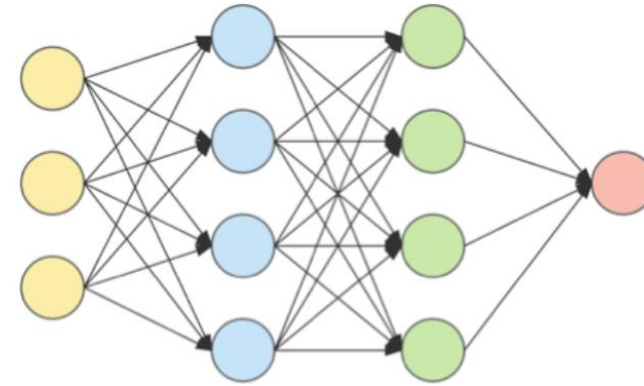
- **Steps**

1. Choose the number of clusters K.
2. Randomly initialize K cluster centers (centroids).
3. Assign each data point to the nearest centroid.
4. Recompute each centroid as the mean of its assigned points.
5. Repeat steps 3–4 until the centroids become stable.



Neural Networks

A neural network is a model made of layers of connected units that learns patterns in data to make predictions.



Steps

- Initialize the network weights randomly.
- Input data passes through the network to produce a prediction (forward pass).
- Compute the loss by comparing prediction with the true label.
- Calculate how each weight affects the loss (backpropagation).
- Update the weights to reduce the loss (gradient descent).
- Repeat until the loss becomes small or stable.